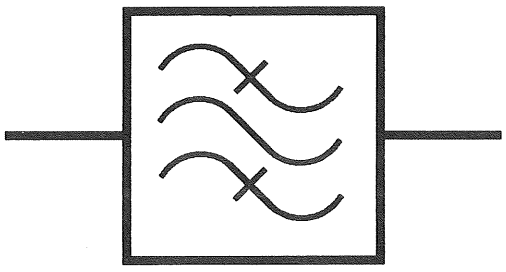
INTRODUCTION TO SIGNALLING

12

REMOTE CONTROL SYSTEMS



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INTRODUCTION

It is a requirement today that a signalling centre covers a large area of control, the limits being perhaps 180 km from the signalling centre. With these big control areas the supervision can be effected more easily as the overview of the whole system allows problems to be more rapidly resolved. However, until the late 1950s there was a limit to the area that could be controlled from one signal box/signalling centre.

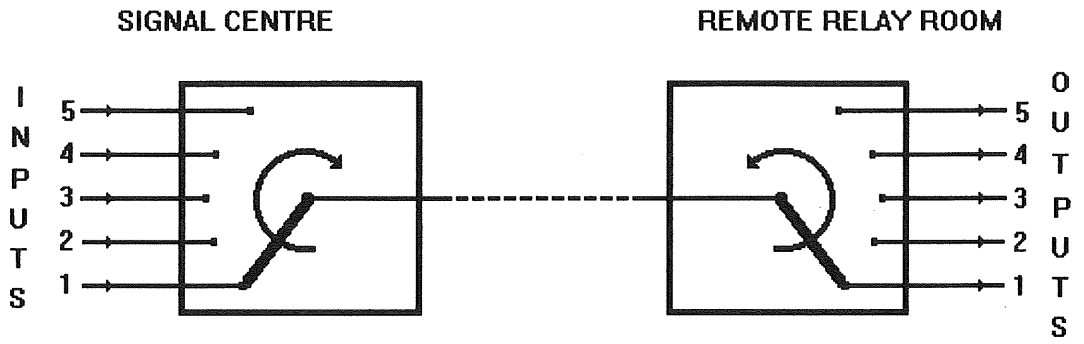
With mechanical boxes the area was limited by the sheer physical effort needed to work points. As a result a limit of a point end being a maximum distance of 200 yards from the box was set. This resulted in large stations needing several boxes to control the area, and this in turn led to inefficiencies unless there was good communications between the signallers.

With relay interlockings there is a different problem - the cost of cable. Consider an area to be controlled 30 km away from the signalling centre. A medium sized interlocking might require 200 controls and 500 indications, each requiring two wires - 1400 wires altogether. This results in a total of 42,000 km of wire, the cost of which would make the whole idea quite unacceptable.

The problem was solved in the late 1950s with the introduction of *remote control systems*, which allow the transmission of many functions along one pair of wires, rather than a pair for each. Two different types of system exist, solving slightly different problems. They are known as *Time Division Multiplex (TDM)* and *Frequency Division Multiplex (FDM)* systems.

THE TIME DIVISION MULTIPLEX SYSTEM

The first TDM system was installed in connection with the Kent Coast Electrification in the late 1950s. The principle behind the idea is shown in the diagram below



The interlocking to control the distant area, the *satellite interlocking*, is housed in a *remote relay room* placed as close as possible to the area. However, the building is unmanned, with no signaller present. This remote relay room is connected to the signalling centre by the TDM system. The signalling centre end, the *transmitter* end, has many inputs, the signaller's controls. The remote relay room end, the *receiver* end, has the same number of outputs, which feed into the interlocking. If any inputs have a voltage applied, by the signaller operating his controls, then the corresponding output must also have a voltage present to make the appropriate change to the interlocking input. A similar system working in the opposite direction would transfer the indications from the satellite interlocking back to the signal box.

The diagram shows a simplified system with just five inputs and consequently five outputs. At the transmitter end a motor driven contact rotates continuously, connected to first input 1, then input 2, then input 3 etc. A wire connected to this arm is connected to a similarly rotating arm at the receiver end. This arm is connected firstly to output 1, then output 2, then output 3 etc. At a particular instant, input 1 will be connected through the arm at the transmitter, through the wire to the remote relay room, and through the receiver arm to output 1. A short time later, input 2 will be connected to output 2, then 3 to 3, 4 to 4 etc. Thus each input is connected in turn to its matching output. After the last input and outputs are connected together, the pattern starts again at input 1. The system is thus able to transmit in practice, many hundreds of functions in sequence down one pair of wires (the second wire of the pair is not shown in the diagram but just acts as the electrical return).

The system does not actually make use of motor driven rotating switches, this concept was used as a simple way of describing the way the system worked. The equivalent switching is carried out electronically with no moving parts. It is important of course for the system to periodically check that the two ends are in synchronism.

If the two ends do become out of synchronism no danger will result as the outputs feed into the interlocking which will ignore all non-safe commands. The TDM system consequently does not need to be a fail-safe device and electronic switching has been used since inception. The very first system, between Sittingbourne and Sheerness in Kent, used a simple valve device, the cold cathode, as the switching device. This was followed over the years by transistors and in time by the ubiquitous microprocessor.

The introduction of TDM systems allowed the control area of a signal box to be greatly increased (and for the term Signalling Centre to be introduced). A large signalling centre might well have in the order of 15 remote relay rooms under its control.

It soon became the practice to send the minimum number of indications back in the TDM system, and then use cheap non-vital relays to organise the actual required indications. These relays were often in pre-wired sets for specific functions, such as all the indications associated with a single point end. This method had other benefits, as it is not practical to send flashing functions through a TDM system. The scanning of the functions would considerably distort the flashing on/off sequence. As a result, two functions had to be sent back for the button light of all controlled signals, one to apply a flashing feed to the lamp, and the other a steady feed as required. Again the non-vital relays would organise the combination of these two functions to the one lamp. In the early 1980s the *Panel Processor* was introduced. This involved the microprocessor equipment at the signal centre also undertaking the function of sorting out the returning indications, rather than using non-vital relays. By now the equipment was housed in the control panel, rather than downstairs in the relay room, which in turn saved many thousands of metres of multicore cabling.

THE FREQUENCY DIVISION MULTIPLEX SYSTEM

On several occasions the need of an a.c. supply of a specific frequency has been mentioned. The dual immune track circuit is an example, as are the Safety Tone and Speed Tone used on the Docklands Light Railway (DLR). Equipment to generate a source of this specific frequency is therefore necessary, as is a device to detect and respond only to that frequency. Considering the dual immune track circuit, the generator would form part of the track feed, whilst the detector would be part of the track relay equipment. On the DLR the generators for the Tones would be trackside, whilst the detectors would be mounted on the vehicles. As all the examples given above have to be fail-safe, the frequency generation and detection must be very precise and not be affected by temperature changes, ageing of equipment or component failure. One way of achieving such frequency-based behaviour is to use *Reed* equipment.

The Reed principle was initially developed in the late 1950s for a Remote Control System, rather than any of the examples mentioned above which came later. The Time Division Multiplex system described above was for the block transfer of many functions between two specific points. There is also a need for a similar Remote Control System which can transfer many distributed trackside functions back to a relay room over one pair of wires. The system devised is known as a *Frequency Division Multiplex (FDM)* System. Each function to be transferred in the system is allocated a specific frequency.

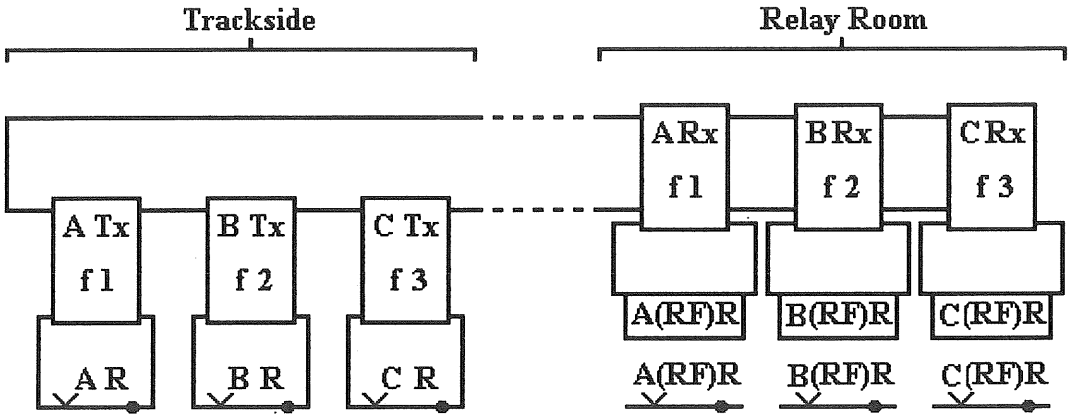
The Principles of a Reed FDM System

Each function to be transmitted requires a generator at the sending end, and a detector at the receiving end, both of the same precise frequency. The generator is known as the *Transmitter* (usually shortened to the telecommunications term *Tx*). The detector is known as the *Receiver* (similarly normally shortened to *Rx*).

Each Tx generates a specific frequency and can be switched on and off by a switch, usually a relay contact.

Each Rx energises an associated relay, but only when it detects a current of the frequency to which it is tuned.

The diagram shows the principle of an FDM Remote Control System, involving, as an example, only three functions A, B and C.

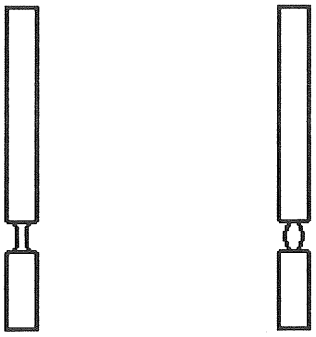


At the left hand end, Function A has a transmitter, A Tx, tuned to frequency f1, switched on or off by a contact of relay AR. When switched on, the frequency f1 is applied to the pair of wires forming the system. At the right hand side is a receiver, A Rx, connected to the same pair of wires and tuned to exactly the same frequency as A Tx. Only if it detects that particular frequency on the pair of wires, the receiver energises a relay, known as a *Reed Follower Relay (RF).* The letters 'RF' are usually incorporated in the relay name, thus 'A(RF)R' as shown. Thus if relay AR is energised, frequency f1 is applied to the line pair and the receiver energises relay A(RF)R. Relay A(RF)R is in effect a repeater of AR, but in a different location. The Tx and Rx for each additional function must both be tuned to the same frequency, which must be different to that used for any other function in the system on that pair of wires.

As currents at different frequencies in the same conductor do not affect each other, each function works completely independently. However it will be appreciated that the frequencies generated must be very stable and never vary. If they did vary, it might be that the frequency of A Tx could change from f1 to f2 and thus energise B Reed Follower Relay causing a wrongside failure.

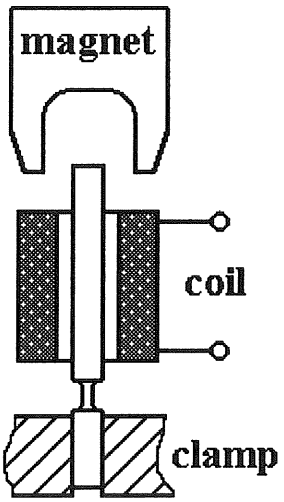
The Reed Filter

The Reed system is based on the simple principle of the tuning fork used by musicians. The frequency emitted by a tuning fork when it vibrates depends solely on the physical dimensions of the fork and is known as the resonant frequency. A similar device is used in the Reed equipment, but of a simpler construction, consisting of a circular rod, or Reed, about 40 mm long and 3 mm diameter.



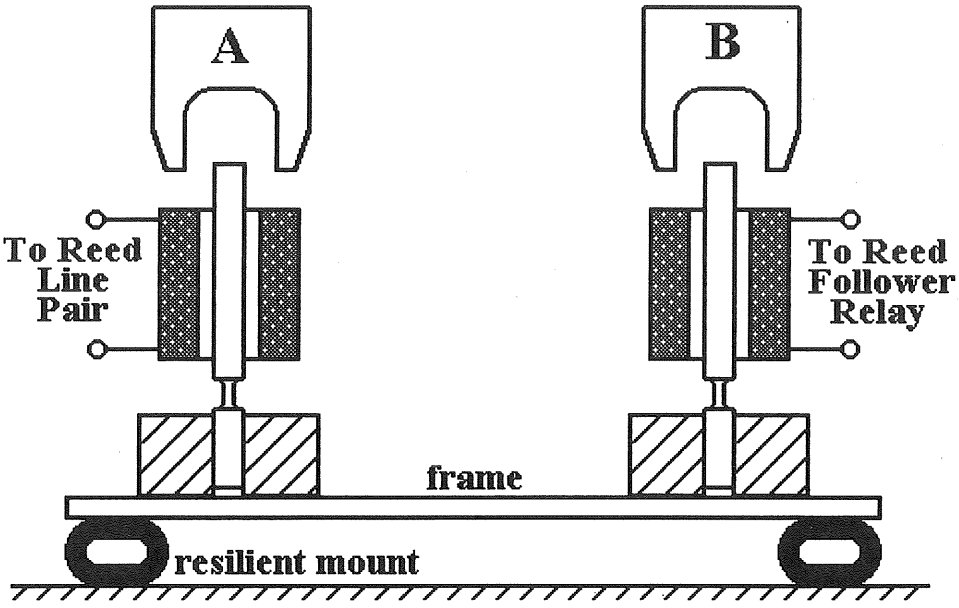
Close to one end the Reed is reduced in size to give a thin section in one plane. It is the thickness of this reduced section that determines the resonant frequency of the top part of the reed when vibrating. This tuning area is filed by hand to achieve the desired resonant frequency. The Reed is made from a special ferromagnetic alloy, carefully chosen so that the resonant frequency, once set, will not be affected by ageing or temperature changes.

The lower end of the Reed is clamped in a brass block, is surmounted by a permanent magnet and enclosed in a electrical coil.



The magnet is positioned so that the top of the Reed, when vibrating (across the page in the diagram above) will be within its magnetic field. The three elements of electromechanics are thus evident - motion (the vibrating Reed), a magnetic field and an electrical conductor (in the coil). The assembly can therefore behave like a 'generator' or as a 'motor' depending on the circumstances. This generator effect will induce a current of the resonant frequency in the coil if, and only if, the Reed is vibrating. The motor effect will result in the Reed vibrating if, and only if, there is a current of the resonant frequency present in the coil. If only non-resonant frequency currents are flowing through the coil, then the Reed will NOT vibrate.

Two of these assemblies bolted onto a resiliently mounted frame make up a *Reed Filter.* Both Reeds must be tuned to the identical resonant frequency, say f1.



Considering the Reed Receiver firstly, the coil of one assembly, A, is connected to the Reed line pair. The other coil, B, is connected via an amplifier and rectifier to the coil of a Reed Follower Relay.

Depending on the number of Transmitters switched on in the system at any moment, a mixture of frequencies will pass through coil A. However, only if the appropriate frequency f1 is present will the Reed actually vibrate. The vibrations will travel through the frame to the B assembly, where its Reed will vibrate in sympathy (as long as it is tuned to precisely the same frequency). As this second Reed is vibrating, a current will be induced in the coil which in turn will energise the Reed Follower Relay. Thus only the presence of a current of frequency f1 on the line will energise the Reed Follower Relay.

Under fault conditions it might be possible to persuade the A Reed to vibrate at a non-resonant frequency if enough power was applied. However, the B Reed would not be forced to vibrate at its non-resonant frequency, so the combination acts as a very accurate filter.

The Transmitter is basically an oscillator, the frequency of which is clamped by applying the well known technique of negative feedback. The Reed filter is in this feedback path and the frequency settles down after one or two seconds to the appropriate Reed resonant frequency. There is thus an inherent delay between the controlling contact of the Transmitter closing, and the Reed Follower being energised. If this is a problem, the transmitter can be wired externally in a different way so that it is continuously oscillating, and the external switching contact switches the frequency directly onto the Reed Line pair with no delay.

Construction

The Tx consists physically of two parts, the filter and the oscillator. The latter is identical for every frequency, so to reduce the amount of spares it is necessary to hold, is in a separate container which clips to any Reed Filter. The combination plugs into a standard Miniature Relay plugboard. However the assembly is slightly larger than a standard relay in the vertical plane, so a special retaining clip is required.

Using similar arguments, the Rx also consists of two parts which clip together, the Filter and the Amplifier.

Every Tx and Rx in the range has a unique pincode to ensure that only equipment of the correct frequency can be plugged in.

The Reed Follower Relay is to the normal Miniature Relay specifications but has a special coil to match the Rxs' output characteristics.

The Non-Vital Reed System

As initially designed, the Reed system was a vital system. However, it was soon realised that a cheaper non-vital system would be desirable. The Tx of the non-vital has a single Reed rather than the double filter, and physically fits in a standard Miniature Relay case. The non-vital Rx still has the double filter, but has a cheap built-in follower relay rather than driving an external vital relay.

The Frequencies

The frequencies chosen for Reed systems are audio frequencies between 350 Hz and 900 Hz. Frequencies in this range were chosen so that Reed circuits could use pairs in normal signalling cables. If higher frequencies had been chosen, then more expensive coaxial cable would have been necessary. In special circumstances, where the cost of especially installing signalling cables may be saved, Reed circuits have been applied in telecommunications cables leased from national telephone systems.

The actual frequencies used are carefully selected from this range of audio frequencies taking into account the frequency of the power supply in the country in which the system is to be installed. The frequencies are chosen to avoid all the harmonics of the national supply, taking into account the allowable variation in that frequency. Reed systems must not be affected by adjacent high voltage electricity authority power lines. Similarly, Reed systems must be completely immune to the effects of the railway traction system even under fault conditions. These factors significantly limit the number of possible *channels* (i.e. frequencies) that may be used in a Reed System.

	MAXIMUM NUMBER OF CHANNELS	
TRACTION TYPE	*VITAL SYSTEM*	*NON-VITAL SYSTEM*
Non Electrified	51	63
50 Hz a.c. or d.c. (rectified from 50 Hz)	16	63
60 Hz a.c. or d.c. (rectified from 60 Hz)	14	63

The band pass of the Reed filter is very narrow (the Q value varies between 500 and 1000, depending on the actual frequency). A deviation of only half a Hertz is sufficient for the filter not to work. It is thus possible to safely use frequencies only a few Hertz apart.

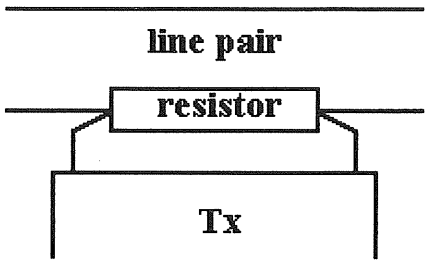
The Practical System

Power Supplies

Both Txs and Rxs need power supply units and a range is available. The smallest for one Rx or two Txs, the biggest for the maximum number of Rxs in a system.

The Transmitters and Receivers

The output of each transmitter is applied across a resistor permanently wired in series with the line.



This ensures that if a Tx is unplugged for maintenance purposes, the line will not be broken and so the rest of the system will carry on working. Additionally, the impedance of the line will not be unduly altered if a Tx is unplugged.

Rxs are connected in parallel across the line, so unplugging one will not affect the working of the others.

Line Amplifiers

In order to maintain adequate signal levels in the line, amplifiers are necessary. These fail-safe amplifiers must be specially designed to ensure that the frequencies passing through them are not changed under any circumstances.

Isolating Transformers

On a.c. electrified lines all trackside relay circuits must be limited in length to 2 km in order to keep induced voltages at acceptable levels. The same rule applies to Reed systems, and the careful positioning of Line Amplifiers will fulfil the requirement. If no amplification is required at the isolation point, a simple isolating transformer is available.

Applications

The common use of non-vital Reed systems would be collecting track circuit information from long stretches of auto signals - a non-vital Reed Indication system. The Txs would be distributed along the trackside in the location cases and all the Rxs would be in the relay room.

Less common would be a Reed Control system, perhaps sending out emergency replacement controls to auto signals and ground frame releases. This would be a vital system, with all the Txs in the relay room and the Rxs distributed along the trackside.

Such systems have information passing only in one direction - *simplex* systems. *Duplex* systems, with information passing in both directions are feasible, as long as the system does not require Line Amplifiers.

It is often possible to save the expense of building a remote relay room if the relevant interlocking is housed in a nearby relay room, and vital Reed systems used to transfer the controls and indications to and from the location cases. Two large schemes, Trent and Saltley, were engineered in this way in the late 1960s. In each contract all the interlockings for the scheme were housed together in the Signal Centre and communication with the outside world was by many hundreds of vital Reed systems.

There are particular immunity problems in areas where both a.c. and d.c. traction systems are in close proximity. An example is between Euston and Watford Junction, where the 'main lines' are electrified on the a.c. system, and the suburban lines (actually always known as the 'd.c. lines') run in parallel. In these circumstances Reed equipment may be used for track circuits, point detection and even for line circuits.

A particularly beneficial use of Reed equipment is for block systems between passing loops on single line railways. The use of Reed can considerably reduce the need for extremely expensive signalling cable between the loops. In some cases signalling cable may be eliminated completely by using a couple of pairs (one in each direction) in the telecommunications cable. In these circumstances signal levels have to kept low to comply with CCITT requirements. In order to ensure safety and overcome signal/noise problems, each function is represented by the simultaneous transmission of two frequencies. Line integrity may also be continuously proved by the transmission of another pair of frequencies.

Economics

When considering the use of Reed a simple calculation is required to balance the cost of cable (including installation and termination) and relays against the equivalent Reed equipment. Each case has a different balance, but in very general terms, a system of eight functions over 2 km long would be an economic proposition.